

Timeline	Activities	Expected Deliverable
Week 1	Collect and study CICIDS2017, UNSW-NB15, MITRE ATT&CK, CVE/NVD, CAPEC, Threat Intelligence Reports, and SOC Playbooks. Perform data preprocessing, feature engineering, train baseline classifiers (Random Forest/XGBoost), and integrate SHAP/LIME for explainability.	Working IDS pipeline with baseline performance metrics and XAI explanations.
Weekend 1	Install and configure Neo4j. Design GraphRAG schema and entity relationships. Prepare cybersecurity corpus for LoRA fine-tuning (ATT&CK, CVE, CTI reports, playbooks, synthetic incident reports).	Graph schema finalized and fine-tuning dataset prepared.
Week 2	Begin LoRA fine-tuning of LLaMA 3.1 8B on university workstations. In parallel, construct the GraphRAG pipeline using Neo4j and ingest ATT&CK, CVE, CAPEC, and CTI data.	Functional GraphRAG system and initial domain-adapted LLM checkpoint.
Weekend 2	Complete pending GraphRAG ingestion tasks, validate retrieval quality, and monitor fine-tuning progress.	Stable GraphRAG retrieval and finalized fine-tuned model.
Week 3	Integrate GraphRAG with the fine-tuned LLM to create the AI Security Analyst. Generate contextual incident reports using classifier outputs, SHAP explanations, and retrieved threat intelligence. Begin implementation of Classifier-LLM Agreement Analysis.	End-to-end TrustSecAI prototype with contextual incident report generation.
Weekend 3	Refine Agreement Analysis module and prepare ATT&CK-based reasoning workflows for Attack Chain Prediction.	Attack-chain prediction design and agreement-analysis framework ready.
Week 4	Implement Attack Chain Prediction Agent using GraphRAG and ATT&CK relationships. Complete Agreement Analysis, perform ablation studies, evaluate system performance, and conduct end-to-end testing.	Fully integrated TrustSecAI framework with all proposed novelties implemented.
Weekend 4	Final documentation, report writing, PPT preparation, result visualization, and demonstration setup.	Final project report, presentation, and demo-ready system.